Quantum Technologies for Autonomous Vehicles

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Introduction

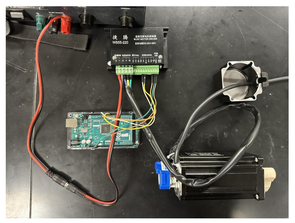
This project emphasizes the properties of quantum entanglement for cryptography (or more commonly known as Quantum Key Distribution) for a more secure form of encrypted communication to relay information between two unmanned vehicles to perform autonomous tasks. This project is currently in the theoretical / prototyping stages as a combination of these technologies is fairly new. Our current method of integrating coarse and fine alignment can be extremely beneficial for a variety of other fields beyon the scope of Quantum Key Distribution. Keeping documentation on how we created our alignment system, we hope to provide an affordable and solid system that other researchers can replicate and use for their own specific projects.

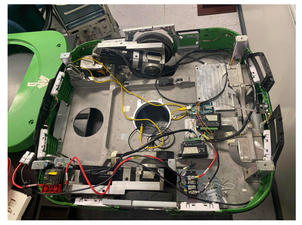
Objective

- Restore IHerb robot so its mobile with commands
- Design new housing for the components and add a lithium battery to power the implemented components
- Program Pixy2 camera to enable autonomous tracking
- Mount the Thorlabs demonstration kit on IHerb and make functional
- Replace demonstration kit with the quantum optics setup
- Program robots to achieve an autonomous task using quantum optics

Materials and Methods

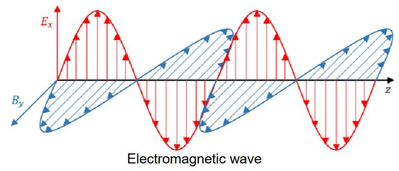
For the restoration of the IHerb robot, we used an Arduino to run 2 brushless motors with 2 BLDC (brushless direct current) motor drivers, and a pixy2cam for motion sensing. We will also be using a PhantomX XL430 for the the laser coarse alignment and a raspberry pi to run it. The robot itself was donated from a bankrupt company courtesy of Dr. Boskovich. We sourced a lot of our code foundation from ChaptGPT and had to make adjustments to it for our purposes. Our main methodology was trial and error, we were given very little info about the IHerb robot and so completed our own research.



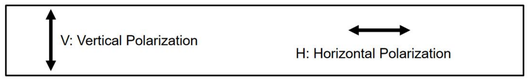


Discussion

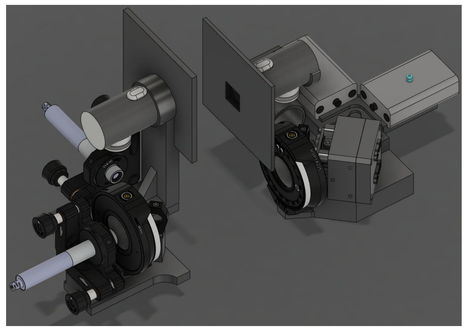
- QKD offers high-security encryption to prevent unauthorized access to confidential information.
- Applications of QKD are vast, but it requires a quantum internet, which may not be widely available when quantum computers become viable.
- Quantum Key Distribution (QKD) uses quantum entanglement to generate random polarized photons.
- Photons' polarization states are unknown until measured by Bob's polarizer.
- Polarization measurements are converted into binary bits to generate the key for cooperative autonomy.
- Alice and Bob publicly compare bases to verify the correctness of the key, keeping measurements with the same basis.
- No-cloning theorem in quantum mechanics ensures secure encryption, preventing replication of photons.
- Intercepted signals reveal errors to detect eavesdropping attempts during communication.

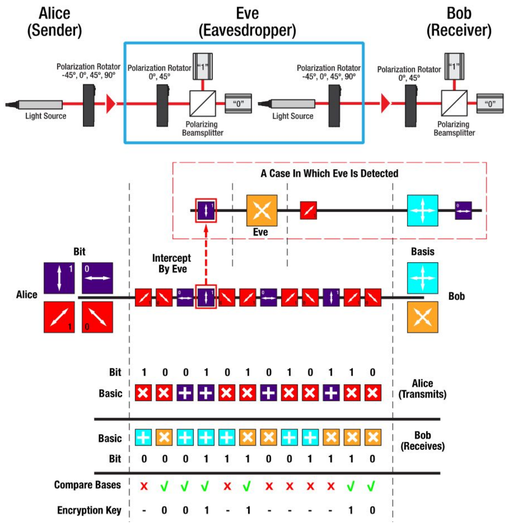


Electromagnetic wave



Khoshnoud, F. Quantum Cryptography and Quantum Entanglement for Engineering. 2021





Thorlab, Quantum Cryptography Demonstration Kit: User Manual. 2020.

Future Work

Minimizing the amount of components used for the robotic vehicle to create a more streamlined approach as opposed to having multiple redundant components (ie. cutting down to one camera instead of two for object tracking). The main issue with this approach would be programing all components to run off a single Raspberry Pi 4B which may be too much for its limited hardware.

We will also have to create an aluminum base plate to mount the beam splitter. This task is composed of measuring components and stabilising the crystal splitter. It should be noted that the slightest variation in the position of the splitter will compromise the connection between the laser and the receiver. Current method of using 3D printed components can lead to slight deformation in the base plate leading to beam diverging from intended target.

Further, we will build a new chassis in order to minimize the UGV's footprint in weight and square footage. This will make it easier to work with and create an easy baseline that can be replicated.

Results

- PixyCam successfully tracks single color blobs and activates motors to move the IHerb.
- Motor successfully controlled by Arduino Code
- 40V Lithium Battery successfully powers up onboard system
- Thorlabs demonstration kit setup
- Laser tracking with coarse and fine alignment is setup and functional on test bench
- Fine alignment which is a critical component, proven to be the most difficult to work with when mounting and prototyping entire system.



Summary and Conclusions

This ongoing project promises new possibilities for incorruptible communication between deployed drones and robotic systems in the field. There are still many hurdles for us to overcome, more specifically with coarse and fine alignment. This portion of the project alone is the key for the entire system to work and has numerous applications beyond quantum key distribution. Further refinement of our current setup must be made and new software methods may get us closer to our end goal. We'd like to acknowledge the help and assistance from Professor Farbod Khoshnoud, Professor Gerald Herder, and Anan Hamdan, our department's technician.

References

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